

ADC-aware PTQ and Post-ADC Low-Rank Correction for Llama

From Dead-Zone Diagnosis to $\text{INT4} \approx \text{INT8}$ PTQ Quality

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Why Analog Inference? Why This Is Hard.

The opportunity: analog MVM

- LLM inference is memory- and compute-bound
- Analog *multiply-accumulate* is 10–100× more energy-efficient vs. digital
- Real benefit only if output precision is sufficient

The bottleneck: ADC

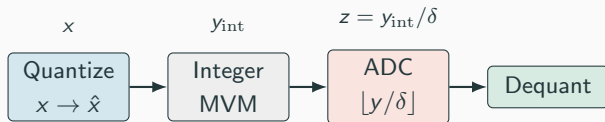
- After each tile MVM, an ADC digitises the analog accumulator
- ADC resolution is *coarse*: $\delta \approx 2016$ (INT8 tile)
- Poor activation distributions $\Rightarrow$ most outputs fall in the **dead zone** ($|y_{\text{int}}| < \delta$)

TODO figure

Digital path (left) vs Analog MVM + ADC path (right). Highlight: ADC is the bottleneck.

Hardware Model & Metrics

One tile of analog MVM



Dead zone: $|y_{\text{int}}| < \delta \Rightarrow z = 0$ (ADC outputs nothing)

$$\delta = \frac{2 \cdot \text{tile_in} \cdot 127^2}{2^{b_a} \cdot k} = \frac{2 \cdot 256 \cdot 127^2}{256 \cdot 16} \approx 2016$$

Key metrics

- $\text{PPL}_{\text{bypass}}$ — PPL with ADC disabled (transform quality proxy)
- PPL_{ADC} — PPL with full ADC pipeline (what matters)

Parameter	Value
Model	Llama-3.2-1B
b_w, b_x	4 or 8 bits
b_a (ADC)	8 bits
Tile size	256
k (parallelism)	16
δ (INT4)	≈ 515
δ (INT8)	≈ 2016
FP baseline PPL	≈ 10.5
Cal. samples	128–1024

Why FlatQuant Transforms Matter for ADC

Without transform: sparse codes

- Per-token scale $s = \max(|x_{\text{tile}}|)/127$
- One large feature gets code ≈ 127 , rest get codes $\approx 1-10$
- Dot product: $y_{\text{int}} \approx 1 \cdot w_{\text{max}} \approx \text{small} \Rightarrow$
dead zone

With FlatQuant transform

- Kronecker-decomposed orthogonal T learned per layer
- After rotation: all features have similar magnitude
- Codes $\approx 50-100$ for all channels
- $y_{\text{int}} \approx 64 \cdot \sum w_i \gg \delta \Rightarrow$ no dead zone

TODO figure

Left: activation histogram before transform (one dominant spike at 127, rest near 0).
Right: after transform (flat, codes 40–100).
Below: resulting y_{int} distribution crossing delta threshold.

First Key Discovery: Dead Zone Was Not Hardware-Fundamental

Early baseline (branch

llama-flatquant-adc)

- $\text{PPL}_{\text{bypass}} = 21.6$, $\text{PPL}_{\text{ADC}} = 28.99$
- $\text{dead_rate} = 81\%$
- PACT, penalties, L1 loss $\Rightarrow$ all failed

Improved baseline (branch pact)

- $\text{PPL}_{\text{bypass}} = 10.0$, $\text{PPL}_{\text{ADC}} = 28.86$
- $\text{dead_rate} = 10.3\%$
- **Hardware unchanged.** Only better FlatQuant training (percentile calibration vs. absmax)

TODO figure

Grouped bar chart: x-axis = {Early baseline, Improved baseline}, two bars per group: $\text{dead_rate} (\%)$ and ADC PPL. Values: $\text{dead_rate } 81\% \rightarrow 10.3\%$; ADC PPL $28.99 \rightarrow 28.86$. Note: ADC PPL barely changes despite $8\times$ dead-rate reduction.

Implication:

INT8: Propagated Calibration Fixes the Real Problem

Two ideas for closing the 10 $\rightarrow$ 29 gap:

1. **Bin-centre loss** $\cos^2(\pi z)$:
nudge y_{int} toward bin centres.
Result: no improvement.
2. **Propagated calibration**:
train layer i on *ADC-quantised* inputs
from layer $i - 1$
(not clean FP inputs).
Result: -47% ADC PPL.

Why propagation works

Standard layer-wise PTQ with FP inputs *underestimates* the reconstruction difficulty at inference, where upstream ADC errors accumulate across layers.

TODO figure

Grouped bar chart (4 methods on x-axis).
Two bars per group: bypass PPL (light) and
ADC PPL (dark). Highlight: center $\approx$
baseline; prop drops ADC PPL from 28.86 to
14.55; prop+center is marginal gain (14.41).
Add FP baseline dashed line at 10.5.

INT4 Pure PTQ: A Stepwise Story

Each step fixes one failure mode:

1. **Baseline INT4 (128 samples):**
 $\text{PPL}_{\text{ADC}} = 2355$ — catastrophic; INT4 overfits on 128 samples.
2. **Propagation (128 samples):**
 $\text{PPL}_{\text{ADC}} = 204$ — better, but bypass *degrades* (transforms over-specialise for corrupted inputs).
3. **Propagation (512+ samples):**
 $\text{PPL}_{\text{ADC}} = 40.6$ — samples are the dominant lever.
4. **Partial propagation $\alpha = 0.5$ (1024 samples):** $\text{PPL}_{\text{ADC}} = 31.2$ — mixed FP+quant loss is better than either alone.
5. **Bounded LET diag + $\alpha = 0.5$:**
 $\text{PPL}_{\text{ADC}} = 27.56$ — diagonal scaling adds expressivity.

TODO figure

Step plot (x = method index 1–6, log y-axis = PPL). Two lines: bypass PPL (blue) and ADC PPL (orange). Annotate each step with the key change. Show FP baseline as dashed line at 10.5 and INT8-prop+center at 14.41. Key numbers: 2355, 204, 40.6, 31.2, 27.56, 27.60.

Mechanistic Insight: MLP and Attention Play Different Roles

Finding (v3/v4 ablation):

- **MLP diag** → best *bypass* PPL, but wide ADC gap
(better transforms, less robust to ADC noise)
- **Attn diag** → does *not* improve bypass, but *closes* the ADC gap
- **Inside MLP diag:** `down_trans` is the key driver,
not `up_gate_trans` (`down_proj` sits right before ADC accumulation)

Staged training exploits this:

1. Stage A (no prop): learn clean MLP diag transforms

TODO figure

Grouped bars: {MLP only, Attn only, Both, Staged}. Two bars per group: bypass PPL and ADC PPL. Highlight: MLP drives bypass, attn drives ADC gap closure.

TODO figure

Grouped bars: {diag up_gate, diag down, diag both}. Show bypass and ADC PPL. Highlight: `down_trans` $\gg$ `up_gate_trans` for ADC quality.

Loss Landscape: Why $\alpha=0.5$ Is the Sweet Spot

Dual-forward mixing loss:

$$\mathcal{L} = (1-\alpha)\mathcal{L}_{\text{FP}} + \alpha\mathcal{L}_{\text{ADC}}$$

- $\alpha=0$: clean FP loss only $\Rightarrow$ flat bypass, wide ADC gap
- $\alpha=1$: full ADC loss $\Rightarrow$ transforms over-specialise for noise, bypass PPL degrades sharply
- $\alpha=0.5$: smooth, wide minimum in both $\mathcal{L}_{\text{bypass}}$ and $\mathcal{L}_{\text{ADC}}$

Visualisation (Li et al. "Filter Normalisation" method):

1. Train FlatQuant to convergence at

TODO figure

3 panels (one per α): 2D contour heatmap of ADC PPL as a function of perturbation (a, b) around θ^* . Color: low PPL = blue (flat/wide), high PPL = red (sharp). Panel 1 ($\alpha=0$): narrow deep valley in bypass dim, flat in ADC dim. Panel 2 ($\alpha=0.5$): wide flat basin in both dims. Panel 3 ($\alpha=1$): very sharp minimum (overfit to ADC noise). See `experiments/loss_landscape.md` for run instructions.

Pure PTQ Hits a Plateau — a Different Tool Is Needed

After all PTQ improvements (v2 through v5):

- Best $\text{PPL}_{\text{ADC}} \approx 27.5\text{--}28.5$ across all INT4 variants
- Adding more calibration samples / stochastic propagation / Hadamard gives < 0.5 PPL gain
- **Root cause:** transform-only PTQ is limited by the coarse ADC floor quantizer ($\delta \approx 515$, only $\sim 25\text{--}40$ bins in the typical z range)

Key realisation

We need a **post-hoc correction** that can learn to model the ADC residual *after* quantisation, without re-training the frozen

TODO figure

Scatter / step plot: all INT4 PTQ variants (v1–v5) on x-axis, ADC PPL on y-axis. Color by version (v1=128s, v2=1024s alpha-sweep, v3=diag, v4=staged, v5=stoch). Draw a horizontal band at 27–28.5 labeled “PTQ plateau”. Draw FP baseline at 10.5 and INT8 prop+center at 14.41 as dashed lines.

Post-ADC Residual LoRA: Breaking the Plateau

Architecture:

$$y = \underbrace{\text{frozen ADC}(x)}_{\text{PTQ path}} + \frac{\alpha}{r} B(A(x_{\mathbb{R}}))$$

- Correction applied **after** ADC output
- Base model frozen in fp16; LoRA adapters in fp32
- Gradients never touch round_ste / ADC clamp $\Rightarrow$ stable
- Trained with LM cross-entropy (+ optional KL teacher), 5 epochs

Pre-ADC LoRA failed

pre_adc_r4_down_o: bypass=123, ADC=105

— catastrophic divergence

TODO figure

Diagram with two branches from x : Top (frozen): $x \rightarrow \text{quantize} \rightarrow \text{tile MVM} \rightarrow \text{ADC floor} \rightarrow y_{\text{base}}$ Bottom (LoRA): $x \rightarrow A \rightarrow B \rightarrow (\text{scaled}) \rightarrow \text{residual}$. Then $y = y_{\text{base}} + \text{residual}$. Red box off to the side: “Pre-ADC LoRA: gradients through round_ste = FAILED”.

ADC-LoRA Ablations: Four Key Findings

1. CE+KL consistently beats CE

- down+o, r4: CE $\rightarrow$ 15.57; CE+KL $\rightarrow$ **14.33** (-1.2)
- all 7, r4: CE $\rightarrow$ 16.68; CE+KL $\rightarrow$ **14.03** (-2.7)
- FP teacher gives information CE alone cannot

2. Rank saturates early

- r=1: 15.90 r=2: 15.60 r=4: 15.57
r=8: 15.24
- Most capacity captured at r=1; higher ranks: diminishing returns

3. First 8 layers beat last 8

- first8: 16.64 last8: 20.08

TODO figure

2×2 grid of small bar charts. Top-left: CE vs CE+KL (down+o and all7). Top-right: rank sweep r=1/2/4/8 (down+o, CE+KL). Bottom-left: first8 vs last8 vs all16 (ADC PPL). Bottom-right: post-ADC vs pre-ADC LoRA — show catastrophic pre-ADC result.

Final Results: INT4 + LoRA $\approx$ FP Baseline

Method	Bits	Wiki2	C4
FP baseline	—	~ 10.5	~ 20.0
INT8 baseline	w8a8	28.86	—
INT8 prop+center	w8a8	14.41	—
INT4 pure PTQ best	w4a4	27.56	—
INT4 staged PTQ	w4a4	27.60	46.40
INT4 + LoRA	w4a4	13.96	23.30

Key messages

- **INT4 + LoRA surpasses best INT8 PTQ on WikiText2**

TODO figure

Grouped bar chart (5 methods on x-axis). Two bars per group: Wiki2 ADC PPL (blue) and C4 ADC PPL (orange). Dashed lines: FP Wiki2 ≈ 10.5 , FP C4 ≈ 20 . Highlight INT4+LoRA as the “winner” bar. Show that C4 improvement is proportionally as large as Wiki2.

Summary of Contributions

1. Dead-zone diagnosis.

Showed that the 81% dead-rate collapse was a *transform-quality problem*, not a hardware limit. Directed all subsequent effort toward transform training, not ADC.

2. ADC-aware INT8 PTQ.

Propagated calibration reduces ADC PPL by 47% ($28.86 \rightarrow 14.41$), approaching FP baseline (10.5).

3. Structured INT4 PTQ.

α -mixed dual-forward loss + bounded diagonal scaling closes the INT4 gap from $\times 153$ to $\times 1.4$ vs. bypass.

4. Mechanistic MLP/Attn insight.

MLP-down_trans drives bypass quality; attn

TODO figure

Flow diagram: "Dead zone (81%)"
 $\xrightarrow{\text{better FlatQuant}}$ "Dead zone (10%)"
 $\xrightarrow{\text{propagated PTQ}}$ "INT8: 14.41" $\xrightarrow{\text{diag} + \alpha}$
"INT4 PTQ: 27.56" $\xrightarrow{\text{plateau}}$ "barrier"
 $\xrightarrow{\text{post-ADC LoRA}}$ "INT4: 13.96" $\xrightarrow{\text{KL} + \text{C4}}$
"Generalisation: 23.30". Connect each arrow with a short label of the contribution.

Backup: Complete INT8 Results (v2)

Experiment	Description	$\text{PPL}_{\text{bypass}}$	PPL_{ADC}	dead_rate
baseline	FlatQuant w8a8	10.01	28.86	10.3%
center mid	bin-centre $\lambda = 0.1$	10.01	28.86	10.3%
prop	propagated calibration	11.01	14.55	10.3%
prop+center	propagated + bin-centre	11.00	14.41	10.3%

Backup: Complete INT4 v2 Alpha-Sweep

Experiment	α	PPL _{bypass}	PPL _{ADC}	gap
baseline_1024	0.0	19.80	56.83	×2.87
prop_full_1024	1.0	30.32	40.20	×1.32
propalpha_075_1024	0.75	27.46	35.92	×1.31
propalpha_05_1024	0.5	24.24	31.22	×1.29
propalpha_025_1024	0.25	21.65	32.35	×1.49
add_diag + $\alpha=0.5$	0.5	20.23	27.56	×1.36
staged_mlp_attn	0.5	18.50	27.60	×1.49

Backup: Complete v7 LoRA Ablation

Experiment	Targets	Loss	Rank	$\text{PPL}_{\text{bypass}}$	PPL_{ADC}
base (no LoRA)	—	—	—	17.89	27.55
r1_down_o	down+o	CE	1	—	15.90
r2_down_o	down+o	CE	2	—	15.60
r4_down_o	down+o	CE	4	18.68	15.57
r8_down_o	down+o	CE	8	—	15.24
r4_down_o_ce_kl	down+o	CE+KL	4	19.54	14.33
r4_down_o_first8	down+o	CE	4	18.42	16.64
r4_down_o_last8	down+o	CE	4	18.62	20.08
pre_adc_r4_down_o	down+o	CE	4	123.34	105.63
r4_all_ce	all 7	CE	4	17.62	16.68
r4_all_ce_kl	all 7	CE+KL	4	19.13	14.03

Backup: v8 Generalisation to C4

Experiment	Targets	Wiki bypass	Wiki ADC	C4 bypass	C4 ADC
base (no LoRA)	—	18.49	27.26	29.41	46.40
rank8_down_o (CE)	down+o	18.51	15.47	30.48	29.39
r4_down_o_ce_kl	down+o	19.05	14.27	29.37	23.88
r4_all_ce_kl	all 7	18.70	13.96	29.10	23.30

Backup: Stochastic Propagation (v5)

Experiment	Mode	$\text{PPL}_{\text{bypass}}$	PPL_{ADC}	gap
stoch_bern (flat)	Bernoulli	19.25	35.38	$\times 1.84$
stoch_beta2 (flat)	Beta(2,2)	22.90	31.40	$\times 1.37$
staged + Bernoulli	Bernoulli	16.94	32.44	$\times 1.92$
staged + Beta(2,2)	Beta(2,2)	17.16	27.82	$\times 1.62$
deterministic $\alpha=0.5$	—	24.24	31.22	$\times 1.29$

Bernoulli gives best bypass but worst ADC gap (hard 0/1 switching provides no ADC signal on FP batches). Beta(2,2) $\approx$ deterministic $\alpha=0.5$ in expectation.

Backup: Hardware Config & Delta Formula

Parameter	Value
b_w (weight bits)	4 / 8
b_x (act. bits)	4 / 8
b_a (ADC bits)	8
<code>mvm_limit</code>	256 / 1024
k (ADC parallelism)	16
δ (mvm=256, INT8)	≈ 2016
δ (mvm=256, INT4)	≈ 515
δ (mvm=1024, INT8)	≈ 2060
FlatQuant epochs	30
FlatQuant LR	0.005
Cal. samples (INT4)	1024
Cal. samples (INT8)	128

Delta formula:

$$\begin{aligned}\delta &= \frac{2 \cdot \text{tile_in} \cdot 127^2}{2^{b_a} \cdot k} \\ &= \frac{2 \cdot 256 \cdot 127^2}{256 \cdot 16} \approx 2016 \quad (\text{INT8})\end{aligned}$$

Number of **ADC bins** in $\pm\sigma_z$:

$$N_{\text{bins}} \approx \frac{2\sigma_z}{\delta/z_{\text{range}}} \approx 15\text{--}30$$

Very coarse — explains why PTQ alone plateaus.